Publishing data sets and code alongside manuscripts

Johanna Ganglbauer, October 2026

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Motivation for data sharing

- Fulfill data sharing requirements
  - Your PI/institution
  - Journal (ES&T)
  - Funding agency (NIEH)
- Open Science
- Improve your research
  - Based on analysis of 46 000 publications in PLOS (Public Library of Science) and BMC (BioMed) Central published before 2015, this study found that **linking manuscripts with publicly available datasets increased the citation index by 25 %.**

The citation advantage of linking publications to research data

Giovanni Colavizza [1,2], Iain Hrynaszkiewicz [3,4], Isla Staden [1,5], Kirstie Whitaker [1,6], Barbara McGillivray [1,6,*]

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Four levels of data availability

1

No additional data available

2

Data available upon request

3

Data available with the paper and its SI

4

Data available in a repository (17% in 2018)

The citation advantage of linking publications to research data

Giovanni Colavizza [1,2], Iain Hrynaszkiewicz [3,4], Isla Staden [1,5], Kirstie Whitaker [1,6], Barbara McGillivray [1,6,*]

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Data Availability Statement

A short statement in the article describing where and how the underlying data can be accessed

Data Availability

Availability of data and software code: All data supporting this study are publicly available through the Harvard Dataverse. (52) The Jupyter notebooks used for data processing and analysis are available on GitHub. (53)

The data that support the findings of this study are available from the corresponding author upon reasonable request.

The data that supports the findings of this study are available in the supporting information of this article.

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Journal Requirements (ES&T)

	ES&T		STEEP	
POLICY LEVEL	**Level 1**	**Level 2**	**Level 3**	**Level 4**
Encourages Data Sharing	X	X	X	X
Encourages Data Availability Statement	X	X	X	X
Requires Data Availability Statement		X	X	X
Requires Data Sharing			X	X
Requires Peer Review of Data				X

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Supporting Information

- PDFs, tables (XLSX), figures,...
- Files hosted alongside the article on the publisher's website
- but difficult to find
- no independent DOI

Publicly available dataset

- Deposited in a repository (e.g., Harvard Dataverse, Zenodo)
- .csv tables with descriptions
- **DOI,** independently citable
- FAIR-aligned (Findable, Accessible, Interoperable, Reusable)

Supporting Information (example)

Supporting Information

Furthering the Capabilities of Diffusive-Gradient Passive Samplers for Per- and Polyfluoroalkyl Substances

136 views | 0 shares | 33 downloads

A	B	
Water Concentrations (ng/L) Measured in Field Demonstration 1	DGT_Measured	DGT
Du_DG_MD_01_W	33.208156411693	4.85
Du_DG_MD_01_W_Dup	30.8298450927285	4.50
Du_DG_MR_02_W	31.3584371626659	4.58
Du_DG_RL_01_W	3.32817315033042	0.48
Du_DG_RL_02_W	2.56022613591848	0.37
Du_DG_RL_02_W_Dup	2.58076899659655	0.37
Du_DG_MD_01_W	25.6231758315652	4.53
Du_DG_MD_01_W_Dup	24.1694287322621	4.28
Du_DG_MR_02_W	19.0988920116933	3.38
Du_DG_BL_01_W	0.474683188659646	0.05

Flow-Tank Conc (Challenge) | S9. Field Demonstration 1 | S10. Field Demonstratio

figshare _ 2 / 2 Share Download

sifile1- pdf file
sifile2- xlsx file

Table (xlsx)

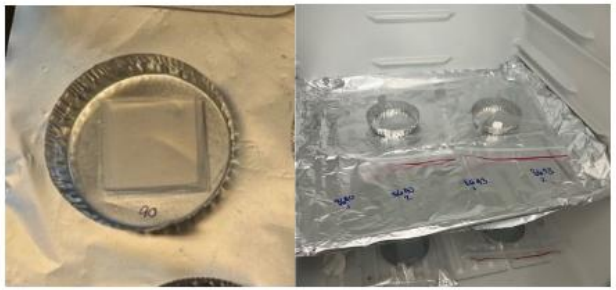
Supporting Information

Furthering the Capabilities of Diffusive-Gradient Passive Samplers for Per- and Polyfluoroalkyl Substances

99 views | 0 shares | 39 downloads



Image S2. Stacked agarose sheets for diffusion experiment (left) which are placed in sealed plastic bags in a temperature chamber for 90 minutes.



Supplemental Text 1.

Diffusion rates were determined by averaging 3 models for diffusion through agarose gel or water based on the individual PFAS molecular weight or volume. The model for diffusion based on Archie's law (Eq. S1) includes parameters adjusting for agarose specific porosity and Archie's law coefficient. The other two models (Eq. S2&S3) describe diffusion through water, but due to the similarity of agarose diffusion (Zhang

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sifile1- pdf file
sifile2- xlsx file

Methods (pdf)

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Publicly available dataset (example)

Diffusive-Gradient Passive Samplers for Per- and Polyfluoroalkyl Substances.

Version 1.1

Jarod Snook, 2025, "Diffusive-Gradient Passive Samplers for Per- and Polyfluoroalkyl Substances.", https://doi.org/10.7910/DVN/V9PKJN, Harvard Dataverse, V1, UNF:6:yTtlqNPWHp5azE3tiLBC/w== [fileUNF]

Cite Dataset Learn about Data Citation Standards.

Access Dataset

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Dataset Metrics

58 Downloads

Description

Per- and polyfluoroalkyl substances (PFAS) are chemical pollutants of growing concern. Due to their ubiquity, persistence in the environment, and potential for toxicity at low environmental concentrations, it is necessary to have convenient and reliable methods to measure PFAS in natural waters. Passive sampling methods (in situ pre-concentration of PFAS) are investigated for monitoring purposes. The following dataset specifically deals with the calibration and application of a diffusive gradient in thin film (DGT) passive sampler, as well as comparisons to previously designed microporous polyethylene tube (MPT) passive samplers and water grab samples. Concentrations are measured with tandem liquid-chromatography mass spectrometry (LC-MS/MS) for all sampling methods. The dataset comprises: (i) a list of 25 PFAS compounds considered for the study at hand, (ii) method detection limits of considered PFAS compounds, (iii) calibration experiments of the DGT passive sampler, as

Read full Description [+]

Subject Chemistry; Earth and Environmental Sciences

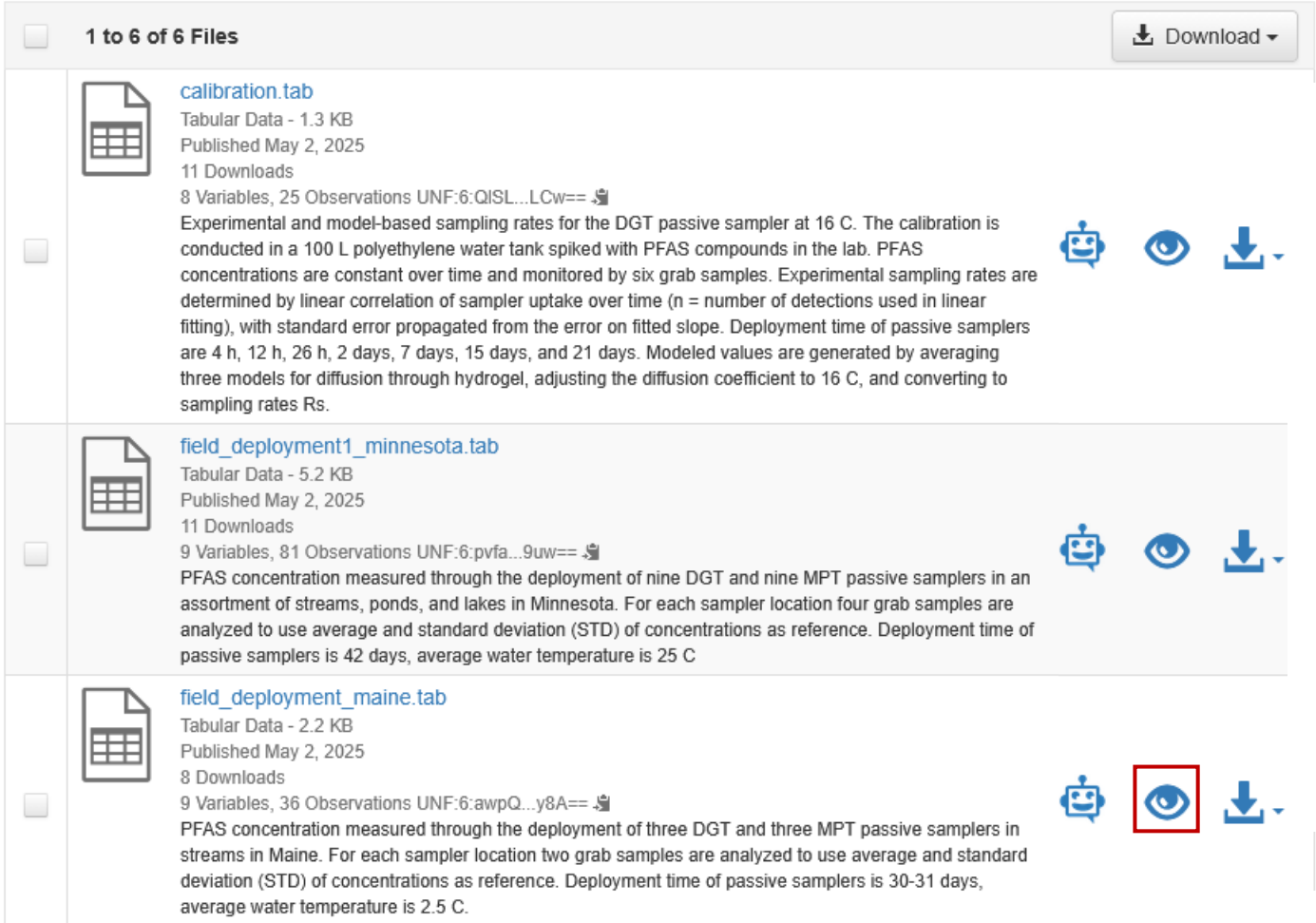
Keyword per- and polyfluoroalkyl substances, Environmental Monitoring, Water Quality, passive sampling, DGT

Related Publication Is Cited By: Furthering the Capabilities of Diffusive-Gradient Passive Samplers for Per- and Polyfluoroalkyl Substances Jarod Snook, Jitka Becanova, Simon Vojta, and Rainer Lohmann Environmental Science & Technology 2025 59 (19), 9744-9753 doi: 10.1021/acs.est.4c14136

License/Data Use Agreement PUBLIC DOMAIN CC0 1.0

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Publicly available dataset (example)



	Location	Site Type	Compound	DGT Concentration in ng/L
1	site_1	Stream	PFBA	2.06
2	site_1	Stream	PFPeA	2.89
3	site_1	Stream	PFHxA	5.65
4	site_1	Stream	PFHpA	15.73
5	site_1	Stream	PFOA	20.62
6	site_1	Stream	PFNA	5.71
7	site_1	Stream	PFBS	0.61
8	site_1	Stream	L-PFHxS	0.26
9	site_1	Stream	Br-PFHxS	0.05
10	site_1	Stream	PFHpS	0.39
11	site_1	Stream	L-PFOS	17.63
12	site_1	Stream	Br-PFOS	11.62
13	site_2	Stream	PFBA	1.93
14	site_2	Stream	PFPeA	2.45
15	site_2	Stream	PFHxA	4.85
16	site_2	Stream	PFHpA	8.86

Publicly available dataset – how to get there?

- Minimal effort: copy of Supporting Information (excel part) -> **Advantages: DOI & findable**
- Best case: data sets contains more detailed information
  - individual results in addition to averages
  - all datapoints used to generate plots
  - more quality assurance/quality control (QA/QC) information
  - …everything needed to reproduce and understand your study

Data Sharing Workflow

1

During review: Deposit dataset (restrict if needed) and get DOI for data availability statement

2

During review: Link dataset DOI in manuscript

3

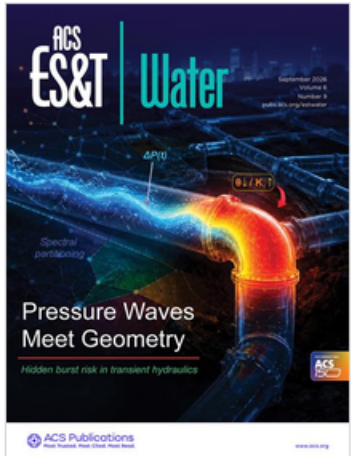
At publication: Open dataset access (when possible, e.g. not for data from human samples)

4

At publication: Reciprocally link dataset → manuscript

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MPT paper: manuscript, dataset and code



Article | August 27, 2026 Open Access

Cross-Study Validation of PFAS Sampling Rates in Microporous Polyethylene Tube Samplers

Johanna Ganglbauer; Matthew Dunn; Jarod Snook; Jitka Becanova; Rainer Lohmann *

+ Author & Article Information

ACS EST Water (2026) 6 (9): 5984–5995.

https://doi.org/10.1021/acsestwater.6c00644 **Article history**

Open PDF

PFAS Sampling Rates in Microporous Polyethylene Tube Samplers

Draft

Ganglbauer, Johanna; Dunn, Matthew; Snook, Jarod; Becanova, Jitka; Lohmann, Rainer, 2025, "PFAS Sampling Rates in Microporous Polyethylene Tube Samplers", https://doi.org/10.7910/DVN/DAH1SO, Harvard Dataverse, DRAFT VERSION, UNF:6:e+Fc1nvlBCqys65YPTejvg== [fileUNF]

Cite Dataset · Learn about Data Citation Standards.

Access Dataset · Publish Dataset · Edit Dataset · Link Dataset · Contact Owner · Share

Description

Dataset Description

This dataset contains PFAS water concentrations measured using grab samples and PFAS accumulated in microporous polyethylene tube (MPT) passive samplers across five field campaigns. These erview of samp The f

HARVARD Dataverse

Read full Description [+]

Subject Chemistry; Engineering; Physics

Keyword per- and polyfluoroalkyl substances, Environmental Monitoring, Water Quality, passive sampling, MPT

Related Publication Is Cited By: Johanna Ganglbauer, Matthew Dunn, Jarod Snook, Jitka Becanova, Rainer Lohmann; Cross-Study Validation of PFAS Sampling Rates in Microporous Polyethylene Tube Samplers. ACS EST Water 2026; https://doi.org/10.1021/acsestwater.6c00644 doi: https://doi.org/10.1021/acsestwater.6c00644

main · 1 Branch · 1 Tag · Go to file · Code

joga4er Update README.md cdbcc1b · 3 weeks ago · 9 Commits

CG_RI	initial commit	9 months ago
JS_DU	initial commit	9 months ago
JS_ME	initial commit	9 months ago
MD_CC	initial commit	9 months ago
MD_WE	initial commit	9 months ago
.gitattributes	initial commit	9 months ago
.gitignore	final improve	2 months ago
LICENSE	initial commit	9 months ago
README.md	Update READ	3 weeks ago
combine_data.ipynb	include Maine challenge deployment	2 months ago
create_plots.ipynb	change line style for model bounds and add to legend	2 months ago
duluth_reevaluation.ipynb	include Maine challenge deployment	2 months ago
sampling_rates.py	final improvements (paper review)	2 months ago

README · MIT license

About

microporous polyethylene tube (MPT) passive sampler

Readme · MIT license · Activity · Custom properties · 0 stars · 0 watching · 0 forks · Report repository

Releases 1

v0.1.0 Latest · 9 months ago

Contributors 1

joga4er

Languages

Jupyter Notebook 86.4% · Python 13.6%

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MPT paper builds upon publicly available datasets from former STEEP trainees

- Christine Gardiner, OSF, 2022: https://osf.io/wzsm9/files/osfstorage
- Jarod Snook, Harvard Dataverse, 2025: https://doi.org/10.7910/DVN/V9PKJN
- Matt Dunn: Supporting Information from two papers

Narragansett Bay

Maine & Duluth

Pawtucket River & Cape Code